Natural Language Processing System Based Interactive Assistive Devices for Person with Disability

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by

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CERTIFICATE

*This is to certify that the work contained in the project entitled "**Natural Language Processing System Based Interactive Assistive Devices for Person with Disability**" is a bonafide work of **Manish M Hanchate (Roll No. 112101002)**, carried out in the Department of Computer Science and Engineering, Indian Institute of Technology Palakkad under my guidance and that it has not been submitted elsewhere for a degree.*

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Abstract

This project aims to develop Natural Language Processing (NLP)-based interactive assistive devices to aid persons with disabilities, particularly those with vision and speaking impairments. The project focuses on two key components: a Text-to-Speech (TTS) system to assist individuals with vision impairments or speech difficulties, and a Voice-Enabled Action Control (VEAC) system to assist those with hearing and vision impairments. Lightweight deep learning networks will be explored for both systems to ensure efficiency, given the resource constraints of portable devices. The systems will be designed and implemented on a Raspberry Pi platform, ensuring accessibility and portability. The expected outcomes include the development of a functional NLP-enabled assistive device prototype, alongside research publications or potential patent filings based on the innovative methodologies used.

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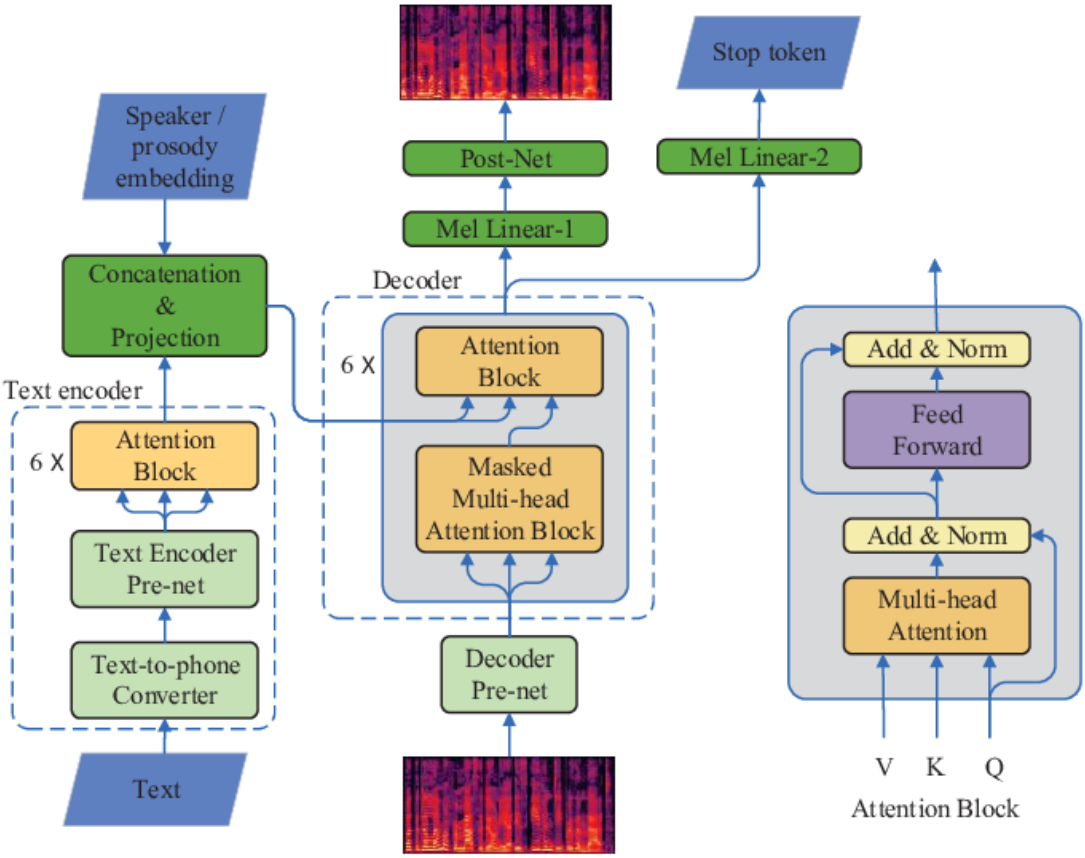

Fig. 1 Block diagram of Transformer TTS

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Chapter 1

Introduction

This chapter introduces the project focused on developing Natural Language Processing (NLP) tools designed for interactive assistive devices aimed at persons with disabilities. The primary goal is to create text-to-speech (TTS) systems to assist people with vision impairments and other disabilities. These systems will be integrated into an application, employing lightweight deep learning networks suitable for the resource constraints of portable devices such as the Raspberry Pi[1, 2].

1.1 Overview

The main aim of this project is to develop natural language processing tools (NLP) with applications in interactive assistive devices for people with disabilities. This project focuses on the development of text-to-speech (TTS) systems to assist people with vision impairment and other problems. These tools will be integrated into an application developed using lightweight deep learning networks to account for the resource constraints of portable devices such as the Raspberry Pi[1, 2, 3].

Recent advances in NLP, particularly Transformer models, have enabled improvements in TTS systems, making them more efficient and effective for assistive technologies. These

models allow for high-quality voice synthesis, which is critical for creating accessible communication solutions for people with disabilities.

TTS systems are crucial in helping people with disabilities, particularly those with vision impairments who rely on these systems to access digital content. In addition, TTS systems can enable voice synthesis for people with speech disabilities, offering them a way to communicate verbally. Using NLP and deep learning, this project aims to enhance accessibility through intuitive voice-controlled interfaces.

1.2 Application of the Project

The developed TTS systems can be deployed on portable devices that are resource-efficient such as Raspberry Pi, making them affordable and accessible. These systems can assist people with vision and speech impairments in various scenarios, such as reading digital text aloud or facilitating verbal communication. The project can also be extended to support other functionalities according to user feedback[3, 7].

1.3 Transformer Model and TTS

The Transformer model is a powerful tool for tasks involving sequences, such as machine translation and speech synthesis. Unlike recurrent neural networks (RNNs), Transformers rely on self-attention mechanisms, allowing for parallel processing of input sequences. The application of Transformers to text-to-speech enables efficient training and high-quality output, making it an ideal choice for this assistive device project[1, 2, 8].

1.4 Existing Methodologies

Existing methodologies for text-to-speech range from traditional concatenative synthesis methods to more advanced approaches based on deep learning. Recent advances in TTS, driven by models like Tacotron and WaveNet, have shown impressive results in generating

natural-sounding speech, which this project aims to incorporate using efficient Transformer-based models[1, 2, 3, 4, 6, 5, 7, 8].

Chapter 2

Review of Prior Works

This chapter reviews the existing literature and developments in text-to-speech (TTS) systems, focusing on traditional methods and the advances brought about by neural networks.

2.1 Overview of Existing TTS Systems

Traditional TTS systems relied on concatenative and parametric methods, which had limitations in the naturalness of speech and the complexity of the system. With the advent of neural networks, systems like Tacotron2 improved speech quality but still faced challenges in efficiency.

2.2 The Role of Transformer Networks

Transformer-based TTS systems address these challenges by replacing RNNs with self-attention layers, improving training speed, and audio output quality. The WaveNet vocoder further enhances the quality by generating natural-sounding speech from mel spectrograms [1, 2, 3, 4, 6, 5, 8].

2.3 Literature Survey

This section provides a survey of key papers in the field of text-to-speech synthesis, outlining their objectives, methodologies, data sets and performance metrics.

2.3.1 Neural Speech Synthesis with Transformer Network

Objective: To provide a comprehensive overview of research advancements in text-to-speech (TTS) synthesis using Transformer networks, focusing on their architecture, training methodologies and performance metrics[1].

Methodology: A systematic literature review was carried out using the following databases: Google Scholar, IEEE Xplore, ACM Digital Library, and SpringerLink. The search terms included "text-to-speech," "Transformer," "neural TTS," "deep learning," and "machine learning." The identified papers were carefully analyzed to extract relevant information about the proposed models, datasets, evaluation metrics, and contributions[1].

Dataset: Commonly used datasets for TTS research include:[1]

- LibriTTS: A large-scale corpus of English read speech
- LJSpeech: A high-quality English speech dataset
- VCTK: A corpus of English speakers from various accents

Technique: Transformer-based TTS models typically follow a similar architecture:[1]

- **Encoder:** Converts the input text into a sequence of embeddings.
- **Decoder:** Generates the target speech waveform or spectrogram.
- **Attention Mechanism:** Aligns the encoder and decoder representations.
- **Vocoder:** Synthesizes the final audio waveform from the generated spectrogram.

Recent advancements in Transformer-based TTS include:[1]

- Multi-head attention: Enables the model to capture different dependencies in the input and output sequences.
- Positional encoding: Incorporates positional information into the input sequence to maintain the order of words.
- Residual connections: Help the model learn complex patterns and avoid vanishing gradients.
- Normalization: Improves the stability and convergence of the training process.

Performance Metrics: Commonly used metrics for evaluating TTS systems include:[1]

- Mean Opinion Score (MOS): A subjective measure based on human ratings.
- Comparison of the mean opinion score (CMOS): A subjective measure that compares two systems.
- Abbreviations Similarity Objective Score (ASOS): An objective measure based on the accuracy of the phoneme.
- Mel-cepstral distortion (MCD): An objective measure based on the difference between the predicted and target mel-cepstral coefficients [1].

2.3.2 Close to Human Quality TTS with Transformer

Objective: The objective of the study is to propose a neural Text-to-Speech (TTS) model based on Tacotron2 and Transformer architecture, with the aim of generating audio samples that are close in quality to human recordings, while allowing parallel training and improving the smoothness of audio prosody[2].

Methodology: The model employs a sequence-to-sequence architecture, where the encoder processes phoneme sequences, and the decoder generates mel spectrograms. Key modifications include using a Transformer for long-distance dependency learning, re-centering

pre-net outputs for positional consistency, and experimenting with different positional encoding methods[2].

Dataset: The model is trained on an internal US English female dataset consisting of high-quality audio recordings paired with their corresponding phoneme sequences[2].

Technique: The approach involves the use of Tacotron2 for mel spectrogram generation, followed by WaveNet as a vocoder to synthesize high-quality audio. Training is carried out using an autoregressive framework with optimizations for faster training times, using the parallel processing capabilities of the Transformer[2].

Performance Metrics: Performance is evaluated using the mean opinion score (MOS) and the comparison mean opinion score (CMOS) for subjective evaluation of audio quality. The study reports MOS values of 4.39 for both the proposed model and Tacotron2, with a slight advantage in CMOS of 0.048 for the new model, indicating improved audio quality. The analysis also includes a detailed evaluation of the reconstruction of the mel spectrogram and the efficiency of training[2].

2.3.3 Tacotron: Towards End-to-End Speech Synthesis

Objective: To develop an end-to-end text-to-speech system that directly converts text to audio, improving naturalness and intelligibility compared to earlier systems[3].

Methodology: Tacotron2 combines an encoder-decoder framework with an attention mechanism to generate mel-spectrograms, which are then converted to waveform using WaveNet[3].

Dataset: The proprietary datasets from Google were used for training and evaluation[3].

Technique: The system consists of two components: a sequence-to-sequence model for generating mel-spectrograms from text, and a WaveNet vocoder for transforming spectrograms into audio waveforms[3].

Performance Metrics: MOS scores were used to evaluate the quality of the synthesized speech, with Tacotron2 achieving high scores, indicating a significant improvement in

naturalness over previous models [3].

2.3.4 Deep Voice 3: Scaling Text-to-Speech with Convolutional Sequence Learning

Objective: The paper introduces a fully convolutional model for text-to-speech synthesis that scales to large datasets and multiple speakers, with faster training and inference times [4].

Methodology: Deep Voice 3 replaces the recurrent networks used in earlier TTS models with convolutional layers to process sequences in parallel, leading to faster training and inference times [4].

Dataset: Trained on the VCTK dataset, which contains recordings of 109 speakers from various accents of English [4].

Technique: A fully convolutional sequence-to-sequence model is used to generate mel-spectrograms from text, followed by a WaveNet vocoder to synthesize audio [4].

Performance Metrics: MOS scores and inference speed were evaluated, showing that Deep Voice 3 achieves competitive speech quality while being faster than RNN-based models [4].

2.3.5 FastSpeech: Fast, Robust and Controllable Text-to-Speech

Objective: To improve the efficiency and controllability of speech synthesis models by introducing a non-autoregressive model that eliminates the need for attention-based alignment [5].

Methodology: FastSpeech uses a feed-forward Transformer with duration prediction to generate speech in parallel, eliminating the sequential nature of autoregressive models [5].

Dataset: The model was trained and evaluated on the LJSpeech dataset and proprietary datasets [5].

Technique: A non-autoregressive model that generates the entire mel-spectrogram in parallel, followed by a vocoder to convert it into audio. Duration prediction is used to handle the alignment between text and audio [5].

Performance Metrics: MOS scores and inference time are reported, with FastSpeech showing comparable or better quality than autoregressive models while being significantly faster [5].

2.3.6 WaveGlow: A Flow-based Generative Network for Speech Synthesis

Objective: The paper presents WaveGlow, a flow-based generative model for producing high-quality speech, combining the efficiency of Glow with the audio quality of WaveNet [6].

Methodology: WaveGlow is a single network trained to directly generate audio waveforms from mel-spectrograms without the need for an autoregressive process [6].

Dataset: WaveGlow was trained on the LJSpeech dataset and additional datasets from NVIDIA for testing and evaluation [6].

Technique: The model uses normalizing flows to generate audio samples in parallel, leading to faster synthesis compared to autoregressive methods like WaveNet [6].

Performance Metrics: MOS scores and inference speed were used to measure performance, with WaveGlow generating high-quality audio with faster inference times compared to WaveNet [6].

2.3.7 FastSpeech 2: Fast and High-Quality End-to-End Text to Speech

Objective: The primary goal of FastSpeech 2 is to address the limitations of existing text-to-speech (TTS) systems such as FastSpeech and Tacotron 2 by improving the quality and speed of speech synthesis. FastSpeech 2 focuses on simplifying the training process

and incorporating variance information (e.g., pitch, energy, and duration) to improve the naturalness and expressiveness of the generated speech [7].

Methodology: FastSpeech 2 trains directly on ground truth targets (mel spectrograms, pitch, energy, and duration) without teacher-student distillation, unlike its predecessor, FastSpeech. The architecture consists of an encoder-decoder structure with additional variance predictors for pitch, energy, and duration. FastSpeech 2s, a variant, further simplifies the process by generating a speech waveform directly from text [7].

Dataset: LJ Speech dataset is used for training and evaluation. In some cases, the authors also use large-scale internal datasets to compare the performance of the model [7].

Technique: FastSpeech 2 builds on the Transformer architecture, replacing the autoregressive approach with a non-autoregressive one. Variance information such as pitch, energy, and duration is explicitly modeled to generate high-quality speech. The continuous wavelet transform (CWT) is utilized for pitch prediction to achieve more accurate pitch contour modeling. FastSpeech 2s extends this by generating speech directly from text using the vocoder [7].

Performance Metrics: MOS (Mean Opinion Score) and CMOS (Comparison Mean Opinion Score) evaluations are conducted to compare FastSpeech 2 and FastSpeech 2s against baseline models such as Tacotron 2 and Transformer TTS. The models are evaluated for both speech quality and inference speed. Additional metrics include pitch contour and energy accuracy, as well as inference time analysis to validate the efficiency of the proposed systems [7].

2.3.8 End-to-End Speech Translation Models

Objective: To investigate the effectiveness of end-to-end models for speech-to-speech translation, particularly comparing the performance of a Transformer-based Transcoder model against existing multi-task approaches [8].

Methodology: The research involved a comparative analysis of various architectures, including RNN and Transformer-based models, evaluating their performance in multiple pairs of languages with differing syntactic similarities [8].

Dataset: The study utilized synthesized data sets that provide parallel speech and text data, ensuring adequate training data to enhance the model's performance across languages [8].

Technique: A Transformer-based Transcoder model was proposed, sequentially addressing speech-to-text, text-to-text and text-to-speech tasks, allowing improved handling of long context information and complex mappings, particularly in syntactically distant languages [8].

Performance Metrics: The performance of the models was evaluated using the Word Error Rate (WER) for ASR, along with the BLEU and METEOR scores to assess translation quality for both text-to-text and speech-to-speech tasks [8].

Chapter 3

Text-to-Speech System Architecture

This chapter outlines the architecture of the proposed text-to-speech (TTS) system, detailing its implementation on the Raspberry Pi and the key components involved.

3.1 Overview

The proposed TTS system will be implemented on the Raspberry Pi using a Transformer model. Key components include:

- **Text-to-Phoneme Conversion:** Reduces ambiguity in pronunciation.
- **Self-Attention Mechanism:** Captures long-range dependencies.
- **WaveNet Vocoder:** Converts mel spectrograms to natural sounding audio[1].

3.2 System Configuration and Setup

The project is currently in the setup phase, with package installations (e.g., TensorFlow, PyTorch), dataset preparation underway, and configuration of the transformer model.

Chapter 4

Application Development

This chapter discusses the development of an application that integrates the text-to-speech (TTS) system designed for individuals with disabilities.

4.1 Overview

As part of the project, I will create an application. This application will integrate the TTS system into a user-friendly interface that assists people with vision impairments and speech difficulties. The application will be compatible with other OS, so it will run efficiently on devices.

4.2 Application Design

The application will be developed in such a way that it can run on limited resources. It will take the text input of the user and process it through the Transformer TTS model to generate real-time speech output. The interface will be simple and accessible, catering to the needs of individuals with disabilities.

Chapter 5

Conclusion and Future Work

This project successfully develops an innovative Natural Language Processing (NLP) system aimed at improving communication and accessibility for individuals with disabilities. Using advanced text-to-speech (TTS) technology and a Voice-Enabled Action Control (VEAC) system, the interactive assistive device offers real-time interaction through voice commands, catering to users with vision and speech impairments.

The use of lightweight deep learning models optimized for portable platforms like Raspberry Pi ensures resource efficiency without compromising performance. The project not only addresses critical communication barriers faced by people with disabilities, but also promotes greater independence and inclusivity in their daily activities. Future work will focus on refining these technologies through user feedback, with the ultimate goal of positively contributing to the quality of life of people with disabilities.

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